Notes: Chen, Hong, and Stein (JFE, 2002)
Breadth of Ownership and Stock Returns

Contents

1 Big picture

- **Question.** Do differences of opinion and short-sale constraints generate predictable cross-sectional stock returns, and can mutual-fund ownership breadth reveal when pessimistic information is not incorporated into prices?
- **Core idea.** If investors disagree and some investors cannot short, pessimists simply stay out. Then price reflects the valuations of optimistic long holders. Low breadth of ownership means few constrained investors are long, which signals that many potential investors are sitting on the sidelines.
- **Main empirical proxy.** Breadth is measured using quarterly mutual-fund holdings. The key variable is $\Delta BREADTH_t$, the change in the fraction of mutual funds that hold the stock.
- **Main prediction.** A reduction in breadth means the short-sale constraint is more tightly binding, so the stock is overpriced relative to fundamentals and should earn lower future returns.
- **Main result.** Stocks in the lowest decile of prior-quarter change in breadth underperform stocks in the highest decile by about 6.38% over the next twelve months. After size, book-to-market, and momentum adjustment, the spread is about 4.95%.
- **Mechanism result.** Breadth is strongly related to momentum. When returns have been poor, many mutual funds exit the stock, but because they cannot short, negative views are only partially impounded in prices.
- **Why this is different from short interest.** Short interest is noisy because low short interest may mean either weak pessimism or high shorting costs. Breadth has a cleaner prediction because it directly measures how many constrained investors are long.
- **Economic takeaway.** Ownership breadth is a market-wide valuation signal. It provides evidence that short-sale constraints matter for prices and that momentum is connected to constrained pessimism.

2 Incremental contribution of the paper

- **From short interest to breadth of ownership.** Earlier empirical work tested short-sale constraints using short interest. Chen, Hong, and Stein argue that short interest is theoretically ambiguous. Their contribution is to propose breadth of ownership as a more robust proxy for the amount of negative information excluded from prices.
- **A direct, broad measure of constrained investor participation.** The paper uses mutual-fund holdings to measure how widely a stock is owned by an important class of investors that typically does not short. This turns a theory about sidelined pessimists into a measurable variable.
- **Link to return predictability.** The paper shows that changes in breadth forecast future returns even after controlling for known predictors such as size, book-to-market, and momentum.
- **Link to momentum.** The paper provides an institutional interpretation of momentum: losers may remain overpriced because constrained investors can sell out but cannot take negative positions.
- **Decomposition of breadth changes.** By separating $\Delta BREADTH$ into new entries IN and exits OUT, the paper studies whether the predictive signal comes from funds entering the stock, funds leaving it, or both.
- **Control for mutual-fund stock-picking.** Because mutual funds may have skill, the paper controls for changes in aggregate mutual-fund holdings $\Delta HOLD$. This helps distinguish breadth from simple net buying by skilled fund managers.
- **Robust empirical design.** The paper combines portfolio sorts, Fama–MacBeth regressions, residual-breadth sorts, size disaggregation, momentum controls, and outlier/quarter robustness checks.

3 Data and measurement

3.1 Mutual-fund holdings

- The core data come from the CDA/Spectrum Mutual Fund Common Stock Holdings/Transactions database.
- The sample covers U.S. mutual-fund equity holdings from 1979Q1 through 1998Q4.
- Mutual funds are well suited for the study because many are legally or contractually unable to short individual stocks.
- The number of funds grows dramatically over the sample, from 582 funds in 1979Q1 to 8,950 funds in 1998Q4.

Detailed interpretation: The changing number of mutual funds makes it impossible to use a simple count of holders. The paper therefore scales by the mutual-fund universe and defines changes using only funds present in both adjacent quarters.

3.2 Breadth variables

For stock i in quarter t,

$$BREADTH_{it} = \frac{\#\{\text{mutual funds long stock } i \text{ in quarter } t\}}{\#\{\text{mutual funds in the database in quarter } t\}}.$$

The change in breadth is measured using the same funds in both quarters:

$$\Delta BREADTH_{it} = \frac{\#\{\text{funds long at } t\} - \#\{\text{funds long at } t-1\}}{\#\{\text{funds present at } t-1 \text{ and } t\}}.$$

It can be decomposed as

$$\Delta BREADTH_{it} = IN_{it} - OUT_{it},$$

where IN is the fraction of continuing funds that open a new position and OUT is the fraction that close a position.

Detailed interpretation: A decline in breadth is not just mutual funds selling some shares. It is a decline in the number of funds willing to hold the stock at all, which is closer to the model's concept of pessimists moving to the sidelines.

3.3 Aggregate mutual-fund holdings and controls

- $HOLD_t$ is total mutual-fund shares held divided by shares outstanding.
- $\Delta HOLD_t$ is the quarter-to-quarter change in aggregate mutual-fund ownership.
- Controls include log size, book-to-market, earnings-to-price, prior-year momentum, and exchange-demeaned turnover.
- Return adjustments use size, book-to-market, and momentum characteristic benchmarks.

Detailed interpretation: $\Delta HOLD$ controls for net mutual-fund demand or stock-picking skill. Breadth is meant to capture dispersion in participation, not just whether mutual funds as a sector are buying or selling.

4 Model and empirical specifications

4.1 Differences of opinion and short-sale constraints

Buyers have valuations V_i uniformly distributed around fundamentals F on $[F - H, F + H]$. Arbitrageurs value the stock at F. If short-sale constraints did not bind for buyers, aggregate demand would be

$$Q_D^U = \frac{1}{2H} \int_{F-H}^{F+H} \gamma_B (V_i - P) dV_i + \gamma_A (F - P),$$

which gives

$$P^U = F - \frac{Q}{\gamma_A + \gamma_B}.$$

With binding short-sale constraints, only buyers whose valuations exceed price hold the stock:

$$Q_D^C = \frac{1}{2H} \int_{P}^{F+H} \gamma_B (V_i - P) dV_i + \gamma_A (F - P).$$

Detailed interpretation: Without short-sale constraints, optimists and pessimists offset each other. With constraints, pessimists drop out and optimists have more influence on price.

4.2 Breadth in the model

The model defines breadth among constrained buyers as

$$B = \min \left\{ \frac{F + H - P^*}{2H}, 1 \right\}.$$

Breadth is one if even the most pessimistic buyers are willing to hold. Breadth approaches zero if only the most optimistic buyers are willing to hold.

Detailed interpretation: A low value of B means that a large mass of constrained investors are absent. This is precisely the information that short interest can miss.

4.3 Theoretical propositions

- **Proposition 1.** As disagreement H rises, breadth falls and expected return $F - P^*$ falls.
- **Proposition 2.** Variation in other parameters such as arbitrageur risk tolerance, buyer risk tolerance, or supply induces a positive correlation between breadth and expected returns.
- **Proposition 3.** Short interest has no robust sign as a return predictor because changes in arbitrageur risk tolerance can make short interest and expected returns move together.

Detailed interpretation: The theory's main empirical implication is not simply "high disagreement predicts low returns." It is that low breadth is a robust indicator of overpricing under short-sale constraints.

4.4 Determinants of breadth

The determinants regressions take the form

$$\Delta BREADTH_{it} = a_t + b_1 \Delta HOLD_{it} + b_2 LOGSIZE_{it} + b_3 BK/MKT_{it} + b_4 MOM12_{it} + b_5 XTURNOVER_{it} + u_{it}.$$

The same specification is repeated with E/P and with IN and OUT as dependent variables.

Detailed interpretation: Positive coefficients on momentum and value-style variables mean that breadth is correlated with other valuation indicators, as predicted if breadth captures constrained pessimism.

4.5 Portfolio tests

Stocks are sorted into deciles by $\Delta BREADTH$ or residual $\Delta BREADTH$, often within size quintiles. The paper reports cumulative returns over one to four quarters for portfolios $P1$ through $P10$ and the spread $P10 - P1$.

Detailed interpretation: If decreasing breadth indicates overpricing, then the low-breadth-change portfolio should underperform the high-breadth-change portfolio.

4.6 Fama–MacBeth regressions

The return-prediction regressions are estimated cross-sectionally each quarter:

$$R_{i,t:t+k} = \alpha_t + \beta_t \Delta BREADTH_{it} + \Gamma_t' X_{it} + \varepsilon_{i,t+k},$$

where X_{it} contains the usual controls. Reported coefficients are time-series averages of quarterly estimates.

Detailed interpretation: The Fama–MacBeth design asks whether breadth predicts returns at the individual-stock level after controlling simultaneously for other cross-sectional predictors.

5 Identification and empirical design

5.1 Core identification problem

The main challenge is to distinguish the short-sale-constraint channel from three alternatives: mutual-fund stock-picking skill, net buying pressure by mutual funds, and standard risk compensation.

5.2 How the design addresses it

- **Breadth versus holdings.** The regressions include $\Delta HOLD$ to separate the number of fund holders from aggregate mutual-fund demand.
- **Residual breadth.** Some portfolio sorts use residual $\Delta BREADTH$ after removing the part predicted by aggregate holdings and other controls.
- **Size disaggregation.** Results are examined across size classes because small stocks have more binding arbitrage limits and more noisiness.
- **Momentum controls.** Since breadth is strongly related to momentum, the paper controls for momentum and also decomposes momentum into shorter-horizon components.
- **IN/OUT decomposition.** The paper checks whether the effect is driven by new fund entry or old fund exit. Both have economically meaningful signs.
- **Future mutual-fund demand.** Robustness tests add future $\Delta HOLD$ to address whether breadth is just predicting future mutual-fund buying pressure.
- **Quarter and outlier checks.** The results are not driven by year-end tax selling, window dressing, or extreme return outliers.

Detailed interpretation: The design is not a natural experiment. Its credibility comes from measurement, controls, and a consistent pattern across tests that lines up with the theory.

6 Section-by-section study guide

- **Introduction.** Sets up Miller's theory and argues that short interest is a flawed empirical proxy.
- **Model.** Shows why breadth has a robust positive relation with expected returns while short interest does not.
- **Data.** Defines breadth, IN, OUT, $HOLD$, and characteristic controls using mutual-fund holdings, CRSP, and Compustat.
- **Determinants of breadth.** Shows that breadth responds to momentum and valuation variables.
- **Return prediction.** Uses portfolio sorts and Fama–MacBeth regressions to show that $\Delta BREADTH$ predicts future returns.
- **Robustness.** Demonstrates that the result survives alternative momentum controls, IN/OUT decomposition, future demand controls, seasonal tests, and outlier truncation.
- **Conclusion.** Argues that short-sale constraints matter for equilibrium prices and that momentum is closely linked to constrained pessimism.

7 Tables and figures

7.1 Table 1: Summary statistics

Read:

- Panel A reports means and standard deviations for breadth, changes in breadth, IN, OUT, aggregate mutual-fund holdings, size, book-to-market, earnings-to-price, turnover, and momentum.
- Average mutual-fund breadth is low: even across all firms, only a small fraction of funds hold any given stock.
- Breadth is much higher for larger stocks than for smaller stocks, which is expected because large stocks are held by more funds.
- Panels B and C show correlations and autocorrelations. Breadth is highly persistent, while $\Delta BREADTH$ is much less persistent.

Detailed interpretation: Table 1 establishes the measurement environment. Mutual funds hold relatively small fractions of the stock universe, so breadth has meaningful cross-sectional variation. The size gradient is important: large stocks are more broadly held, so size controls and size-stratified sorts are necessary.

Economic message: Breadth is not just another ownership share. It captures how many constrained investors are willing to be long, which is the key theoretical object.

Table 1
Summary statistics

The sample includes stocks from the NYSE, AMEX, and NASDAQ between 1979 and 1998. $\mathrm{BREADTH}_t$ is the fraction of all mutual funds long the stock at the end of quarter t. $\Delta\mathrm{BREADTH}_t$ is the change in breadth of ownership from the end of quarter $t-1$ to quarter t. IN_t is the fraction of mutual funds in the sample at both quarters $t-1$ and t that have established a new position in a stock at quarter t. OUT_t is the fraction of mutual funds that have completely removed an existing position in a stock at quarter t. $\Delta\mathrm{HOLD}_t$ is the fraction of shares outstanding of a stock held by mutual funds at the end of quarter t. $\Delta\mathrm{HOLD}_t$ is the change in the fraction of shares held by mutual funds from the end of quarter $t-1$ to quarter t. $\mathrm{LOGSIZE}_t$ is the log of market capitalization measured at the end of quarter t. $\mathrm{BK/MKT}_t$ is the most recently available observation of book-to-market ratio at the end of quarter t. E/P_t is past year's earnings per share divided by the price at the end of quarter t. NYSE/AMEX $\mathrm{TURNOVER}_t$ is the share turnover in quarter t among stocks listed on NYSE and AMEX. NASDAQ $\mathrm{TURNOVER}_t$ is the share turnover in quarter t of stocks listed on NASDAQ. $\mathrm{XTURNOVER}_t$ is share turnover demeaned within each quarter by the average turnover for the firm's exchange (either NYSE/AMEX or NASDAQ). $\mathrm{MOM12}_t$ is the raw return in the twelve months up to quarter t. Size quintiles are determined using NYSE breakpoints.

Panel A: Means and standard deviations

		All firms	Quintiles 2–5 firms	Quintile 5 (largest) firms	Quintile 4 firms	Quintile 3 firms	Quintile 2 firms	Quintile 1 (smallest) firms
$\mathrm{BREADTH}_t$	Mean	1.29%	2.30%	7.09%	2.56%	1.43%	0.76%	0.25%
	Std. dev.	2.46%	3.12%	5.10%	1.42%	0.93%	0.55%	0.25%
$\Delta\mathrm{BREADTH}_t$	Mean	0.00%	0.00%	−0.02%	−0.01%	0.00%	0.00%	−0.01%
	Std. dev.	0.34%	0.46%	0.91%	0.46%	0.31%	0.20%	0.10%
IN_t	Mean	0.15%	0.27%	0.71%	0.32%	0.20%	0.10%	0.03%
	Std. dev.	0.37%	0.49%	0.87%	0.42%	0.28%	0.18%	0.08%
OUT_t	Mean	0.16%	0.27%	0.73%	0.32%	0.19%	0.10%	0.03%
	Std. dev.	0.30%	0.37%	0.57%	0.29%	0.21%	0.14%	0.07%
HOLD_t	Mean	8.58%	10.90%	10.80%	12.33%	11.39%	9.88%	6.19%
	Std. dev.	8.62%	9.26%	8.37%	9.96%	9.80%	8.73%	7.17%
$\Delta\mathrm{HOLD}_t$	Mean	0.12%	0.19%	0.16%	0.16%	0.22%	0.19%	0.05%
	Std. dev.	2.89%	2.55%	1.63%	2.38%	2.61%	2.90%	3.20%
$\mathrm{LOGSIZE}_t$	Mean	5.049	6.424	8.578	7.169	6.206	5.295	3.638
	Std. dev.	1.818	1.341	0.999	0.612	0.565	0.568	0.961
$\mathrm{BK/MKT}_t$	Mean	0.722	0.664	0.620	0.658	0.649	0.697	0.780
	Std. dev.	2.238	0.568	0.463	0.487	0.502	0.675	3.132
E/P_t	Mean	−0.194	0.038	0.068	0.058	0.043	0.005	−0.411
	Std. dev.	0.954	0.127	0.052	0.074	0.107	0.200	1.511
NYSE/AMEX	Mean	15.8%	17.6%	17.7%	18.8%	17.7%	16.3%	12.4%
$\mathrm{TURNOVER}_t$	Std. dev.	15.7%	16.3%	13.8%	16.2%	17.4%	17.2%	14.0%
NASDAQ $\mathrm{TURNOVER}_t$	Mean	29.8%	37.5%	44.0%	42.8%	39.4%	34.7%	25.2%
	Std. dev.	35.6%	42.9%	48.0%	49.5%	45.5%	38.9%	29.6%
$\mathrm{MOM12}_t$	Mean	20.2%	24.3%	24.0%	23.5%	24.2%	24.8%	15.7%
	Std. dev.	62.9%	51.7%	36.2%	44.2%	50.7%	61.1%	72.8%
	No. of obs.	204,829	103,747	16,740	19,927	27,216	39,864	101,082

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(a) Table 1, Panel A: Means and standard deviations

Panel B: Contemporaneous correlations (using only firms above 20th percentile in size)

	$\mathrm{BREADTH}_t$	$\Delta\mathrm{BREADTH}_t$	IN_t	OUT_t	HOLD_t	$\Delta\mathrm{HOLD}_t$	$\mathrm{LOGSIZE}_t$	$\mathrm{BK/MKT}_t$	E/P_t	$\mathrm{XTURNOVER}_t$	$\mathrm{MOM12}_t$
$\mathrm{BREADTH}_t$		0.056	0.781	0.718	0.080	0.010	0.691	−0.062	0.084	0.090	−0.011
$\Delta\mathrm{BREADTH}_t$			0.217	−0.336	0.026	0.185	0.014	−0.025	0.013	0.002	0.167
IN_t				0.632	0.127	0.106	0.596	−0.082	0.062	0.213	0.096
OUT_t					0.059	0.122	0.533	−0.043	0.047	0.231	−0.079
HOLD_t						0.179	0.231	−0.207	−0.022	0.293	0.071
$\Delta\mathrm{HOLD}_t$							0.023	−0.023	0.023	−0.019	0.128
$\mathrm{LOGSIZE}_t$								−0.160	0.092	0.027	0.069
$\mathrm{BK/MKT}_t$									0.123	−0.100	−0.113
E/P_t										−0.092	0.001
$\mathrm{XTURNOVER}_t$											0.149
$\mathrm{MOM12}_t$											

Panel C: Autocorrelations and cross-autocorrelations (using only firms above 20th percentile in size)

	$\mathrm{BREADTH}_{t-1}$	$\Delta\mathrm{BREADTH}_{t-1}$	IN_{t-1}	OUT_{t-1}	HOLD_{t-1}	$\Delta\mathrm{HOLD}_{t-1}$	$\mathrm{LOGSIZE}_{t-1}$	$\mathrm{BK/MKT}_{t-1}$	E/P_{t-1}	$\mathrm{XTURNOVER}_{t-1}$	$\mathrm{MOM12}_{t-1}$
$\mathrm{BREADTH}_t$	0.989	0.061	0.779	0.712	0.078	0.009	0.697	−0.063	0.079	0.090	−0.019
$\Delta\mathrm{BREADTH}_t$	−0.092	0.028	−0.017	−0.100	−0.025	0.046	−0.008	−0.012	0.019	−0.009	0.125
IN_t	0.751	0.088	0.714	0.630	0.097	0.020	0.587	−0.080	0.053	0.200	0.072
OUT_t	0.766	−0.006	0.695	0.718	0.092	−0.026	0.548	−0.057	0.057	0.232	−0.037
HOLD_t	0.076	0.038	0.126	0.054	0.962	0.168	0.229	−0.206	−0.033	0.286	0.090
$\Delta\mathrm{HOLD}_t$	−0.016	0.028	0.004	−0.026	−0.100	−0.040	0.010	−0.013	0.018	−0.020	0.075
$\mathrm{LOGSIZE}_t$	0.690	0.019	0.591	0.530	0.226	0.015	0.988	−0.154	0.087	0.022	0.016
$\mathrm{BK/MKT}_t$	−0.060	−0.041	−0.090	−0.039	−0.201	−0.034	−0.164	0.907	0.107	−0.099	−0.161
E/P_t	0.074	−0.003	0.063	0.042	−0.037	−0.001	0.090	0.143	0.798	−0.079	0.010
$\mathrm{XTURNOVER}_t$	0.092	0.024	0.205	0.197	0.304	0.033	0.033	−0.099	−0.088	0.849	0.180
$\mathrm{MOM12}_t$	−0.035	0.147	0.072	−0.096	0.038	0.134	0.003	−0.049	0.028	0.127	0.711

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(b) Table 1, Panels B-C: Correlations and autocorrelations

7.2 Table 2 and Table 3: Determinants of breadth, IN, and OUT

Read:

- Table 2 shows that $\Delta BREADTH$ is strongly positively related to $\Delta HOLD$, as expected, but also strongly positively related to momentum.
- The momentum coefficient is economically large and statistically strong in both the book-to-market and earnings-to-price specifications.
- Table 3 decomposes $\Delta BREADTH$ into IN and OUT. Positive $\Delta BREADTH$ comes from more new fund entries and fewer exits.
- The momentum relation appears in both the entry and exit margins, which means past winners attract new fund holders and lose fewer existing holders.

Detailed interpretation: These tables connect the theory to known predictors of returns. If momentum is a non-risk predictor of expected returns, then the theory predicts that breadth should move with momentum. The strong momentum relation suggests that momentum may partly reflect the inability of pessimistic constrained investors to short losers.

Economic message: The breadth signal is not random. It moves systematically with valuation indicators and fund trading behavior, especially momentum.

Table 2
Determinants of ΔBREADTH
The sample includes stocks from the NYSE, AMEX, and NASDAQ between 1979 and 1998 with a market capitalization above the 20th percentile using NYSE breakpoints. The dependent variable is $\Delta BREADTH_t$, the change in the breadth of ownership for a stock in quarter t. $\Delta HOLD_t$ is the change in aggregate mutual fund holdings of a stock in quarter t. $LOGSIZE_t$ is the log of market capitalization at the end of quarter t. BK/MKT_t is the most recently available observation of book-to-market ratio at the end of quarter t. E/P_t is past year's earnings per share divided by the price at the end of quarter t. MOM12 is the raw return from the beginning of quarter $t-3$ to the end of quarter t. $XTURNOVER_t$ is share turnover demeaned within each quarter by the average turnover for the firm's exchange (either NYSE/AMEX or NASDAQ). Size quintiles are determined using NYSE breakpoints. The coefficients reported in the table are time-series means of the coefficients from cross-sectional regressors run every quarter (i.e., Fama-MacBeth (1973) coefficients). The coefficients for the full sample are averages of the size subsample coefficients. t-statistics, which are in parentheses, are adjusted for serial correlation and heteroskedasticity.

Panel A: Specification including BK/MKT_t

	$\Delta HOLD_t$	$LOGSIZE_t$	BK/MKT_t	MOM12	$XTURNOVER_t$	Average R^2 (%)	No. of qtrs.
Size quintile 2	0.0504	0.0002	0.0000	0.0008	−0.0001	33.3	79
	(7.86)	(2.66)	(0.34)	(13.46)	(0.41)		
Size quintile 3	0.0729	0.0005	−0.0001	0.0015	−0.0006	34.8	79
	(7.80)	(3.75)	(0.98)	(12.51)	(2.65)		
Size quintile 4	0.1164	0.0006	−0.0001	0.0025	−0.0004	33.1	79
	(7.40)	(2.73)	(1.22)	(11.45)	(1.22)		
Size quintile 5	0.2560	−0.0001	−0.0004	0.0068	−0.0005	32.0	79
	(7.65)	(0.20)	(2.02)	(11.29)	(0.44)		
Full sample	0.1239	0.0003	−0.0002	0.0029	−0.0004	33.1	79
	(7.97)	(2.05)	(2.06)	(13.04)	(1.12)		

Panel B: Specification including E/P_t

	$\Delta HOLD_t$	$LOGSIZE_t$	E/P_t	MOM12	$XTURNOVER_t$	Average R^2 (%)	No. of qtrs.
Size quintile 2	0.0503	0.0002	0.0003	0.0008	0.0000	34.8	79
	(7.86)	(2.64)	(2.74)	(12.88)	(0.15)		
Size quintile 3	0.0729	0.0005	0.0005	0.0015	−0.0006	33.1	79
	(7.79)	(3.64)	(2.94)	(12.41)	(2.44)		
Size quintile 4	0.1159	0.0006	0.0011	0.0025	−0.0003	31.8	79
	(7.42)	(2.74)	(2.73)	(11.41)	(0.91)		
Size quintile 5	0.2575	−0.0001	0.0018	0.0068	−0.0005	32.9	79
	(7.68)	(0.16)	(1.69)	(11.73)	(0.42)		
Full sample	0.1242	0.0003	0.0009	0.0029	−0.0003	33.2	79
	(7.98)	(2.07)	(3.25)	(13.33)	(0.99)		

argument is closely related to the observation that the bulk of profits in momentum strategies appear to come from the short side of the trade (Hong et al., 2000). It also

(a) Table 2: Determinants of $\Delta BREADTH$

Table 3
Determinants of IN and OUT
The sample includes stocks from the NYSE, AMEX, and NASDAQ between 1979 and 1998 with a market capitalization above the 20th percentile using NYSE breakpoints. The dependent variables are IN_t and OUT_t. IN_t is the fraction of mutual funds in the sample at both quarters $t-1$ and t that have established a new position in a stock at quarter t. OUT_t is the fraction of mutual funds that have completely removed an existing position in a stock at quarter t. $\Delta HOLD_t$ is the change in aggregate mutual fund holdings of a stock in quarter t. $LOGSIZE_t$ is the log of market capitalization at the end of quarter t. BK/MKT_t is the most recently available observation of book-to-market ratio at the end of quarter t. MOM12 is the raw return from the beginning of quarter $t-3$ to the end of quarter t. $XTURNOVER_t$ is share turnover demeaned within each quarter by the average turnover for the firm's exchange (either NYSE/AMEX or NASDAQ). The coefficients are the averages of size subsample coefficients, where the size subsample coefficients are time-series means of the coefficients from cross-sectional regressors run every quarter (i.e., Fama-MacBeth (1973) coefficients) within each size quintile. t-statistics, which are in parentheses, are adjusted for serial correlation and heteroskedasticity.

	$\Delta HOLD_t$	$LOGSIZE_t$	BK/MKT_t	MOM12	$XTURNOVER_t$	Average R^2 (%)	No. of qtrs.
Panel A: Determinants of IN							
Size quintile 2	0.0260	0.0009	−0.0001	0.0001	0.0033	36.3	79
	(7.56)	(9.05)	(2.38)	(3.58)	(8.03)		
Size quintile 3	0.0370	0.0013	−0.0004	0.0004	0.0052	38.0	79
	(7.23)	(15.53)	(7.32)	(3.73)	(8.45)		
Size quintile 4	0.0596	0.0021	−0.0004	0.0005	0.0086	38.2	79
	(7.38)	(15.69)	(5.59)	(3.14)	(8.35)		
Size quintile 5	0.1279	0.0050	−0.0013	0.0007	0.0201	55.8	79
	(7.26)	(23.26)	(7.01)	(2.12)	(8.14)		
Full sample	0.0626	0.0024	−0.0005	0.0004	0.0093	42.1	79
	(8.01)	(24.08)	(8.50)	(3.32)	(8.87)		
Panel B: Determinants of OUT							
Size quintile 2	−0.0217	0.0006	−0.0001	−0.0006	0.0033	33.9	79
	(6.82)	(11.72)	(3.79)	(11.39)	(7.94)		
Size quintile 3	−0.0328	0.0007	−0.0003	−0.0010	0.0056	39.4	79
	(7.03)	(16.30)	(7.66)	(12.58)	(8.49)		
Size quintile 4	−0.0512	0.0013	−0.0003	−0.0018	0.0089	39.4	79
	(6.03)	(15.96)	(4.51)	(14.32)	(8.87)		
Size quintile 5	−0.1216	0.0041	−0.0008	−0.0053	0.0199	53.0	79
	(6.98)	(20.00)	(4.46)	(15.75)	(9.12)		
Full sample	−0.0568	0.0017	−0.0004	−0.0022	0.0094	41.4	79
	(7.24)	(23.17)	(6.79)	(17.93)	(9.45)		

portfolio sorts. Consider first Panel A of Table 4. Here our aim is to forecast raw

(b) Table 3: Determinants of IN and OUT

7.3 Table 4: Returns to portfolio strategies based on $\Delta BREADTH$

Read:

- Panel A reports raw returns. Stocks with the largest increases in breadth outperform those with the largest decreases in breadth over one, two, three, and four quarters.
- The four-quarter raw spread between the top and bottom breadth-change deciles is large and statistically significant.
- Panel B shows that size and book-to-market adjustment does not eliminate the pattern.
- Panel C further adjusts for momentum, and the return spread becomes smaller but remains meaningful.

Detailed interpretation: Table 4 is the main portfolio evidence. The raw returns are striking, and the characteristic adjustments show that the effect is not merely a size or value anomaly. Momentum explains part of the effect, which is expected given the theory, but breadth retains incremental predictive power.

Economic message: Declines in breadth forecast underperformance because they reveal overpricing caused by constrained pessimists sitting out of the market.

Table 4

Returns to portfolio strategies based on ΔBREADTH

The sample includes stocks from NYSE/AMEX and NASDAQ between 1979 and 1998 with a market capitalization above the 20th percentile using NYSE breakpoints. In each quarter t, stocks are ranked (into deciles) relative to other stocks in their size quintile on the basis of their change in breadth, $\Delta BREADTH_t$. Then for stocks in similar deciles of $\Delta BREADTH_t$ across the size quintiles, an equal-weighted portfolio is formed and the performance is tracked over four quarters. In each quarter t, for stocks in size quintiles, residual $\Delta BREADTH_t$ is formed by regressing $\Delta BREADTH_t$ on $\Delta HOLD_t$, the change in aggregate holdings of mutual funds in that quarter. Stocks are ranked (into deciles) relative to other stocks in their size quintile on the basis of residual $\Delta BREADTH_t$. Then for stocks in similar deciles of residual $\Delta BREADTH_t$, an equal-weighted portfolio is formed and the performance is tracked over four quarters. This table reports the average returns of the portfolios in each decile of the two sorts on $\Delta BREADTH_t$ and residual $\Delta BREADTH_t$ along with the difference in the returns of portfolios in deciles 10 and 1, P10−P1. Panels A, B, and C present these results using raw returns, size/book-to-market adjusted returns, and size/book-to-market/momentum adjusted returns, respectively. t-statistics, which are in parentheses, are adjusted for serial-correlation using a Newey-West estimator with lags of up to four quarters.

Cumulative returns after		Sort on ΔBREADTH				Sort on residual ΔBREADTH			
		1 Quarter	2 Quarters	3 Quarters	4 Quarters	1 Quarter	2 Quarters	3 Quarters	4 Quarters
Panel A: Raw returns									
Decile	1	2.93%	5.83%	9.27%	13.48%	2.99%	6.03%	9.53%	14.07%
		(3.07)	(3.49)	(3.94)	(4.47)	(3.08)	(3.56)	(4.03)	(4.60)
	2	3.35%	7.19%	11.50%	16.52%	3.84%	7.87%	11.96%	16.65%
		(3.44)	(4.18)	(4.78)	(5.27)	(4.04)	(4.54)	(4.96)	(5.47)
	3	3.80%	8.02%	12.04%	16.41%	3.64%	8.04%	12.55%	17.45%
		(4.27)	(4.83)	(5.10)	(5.60)	(3.99)	(4.85)	(5.17)	(5.77)
	4	4.11%	8.56%	12.80%	17.69%	3.89%	8.29%	12.79%	17.69%
		(4.83)	(5.30)	(5.44)	(5.93)	(4.61)	(5.28)	(5.47)	(5.95)
	5	3.65%	8.04%	12.67%	17.95%	3.91%	8.21%	12.39%	17.55%
		(4.43)	(5.00)	(5.46)	(6.04)	(4.38)	(4.94)	(5.29)	(5.83)
	6	4.10%	8.41%	12.90%	17.94%	3.85%	8.25%	12.84%	17.49%
		(4.42)	(4.86)	(5.15)	(5.76)	(4.53)	(5.05)	(5.44)	(6.11)
	7	3.83%	8.21%	12.53%	17.29%	4.03%	8.33%	12.62%	17.13%
		(4.35)	(4.99)	(5.32)	(5.79)	(4.47)	(4.88)	(5.34)	(5.71)
	8	4.45%	9.08%	13.63%	18.33%	4.12%	8.42%	13.08%	17.88%
		(4.56)	(4.98)	(5.49)	(5.93)	(4.28)	(4.61)	(5.11)	(5.51)
	9	4.39%	8.89%	13.76%	18.57%	4.34%	8.72%	13.27%	18.00%
		(4.57)	(4.76)	(5.16)	(5.70)	(4.74)	(4.73)	(5.17)	(5.65)

(a) Table 4, Part 1: Raw returns

Table 4 (*continued*)

Cumulative returns after		Sort on ΔBREADTH				Sort on residual ΔBREADTH			
		1 Quarter	2 Quarters	3 Quarters	4 Quarters	1 Quarter	2 Quarters	3 Quarters	4 Quarters
	10	4.95%	9.66%	14.78%	19.86%	4.94%	9.80%	14.96%	20.32%
		(4.47)	(4.68)	(5.04)	(5.33)	(4.43)	(4.85)	(5.17)	(5.54)
	P10−P1	2.02%	3.82%	5.51%	6.38%	1.96%	3.77%	5.43%	6.25%
		(3.96)	(4.66)	(4.52)	(4.08)	(3.97)	(5.01)	(5.24)	(4.67)
Panel B: Size and book-to-market-adjusted returns									
Decile	1	−0.99%	−2.24%	−3.03%	−3.52%	−0.95%	−2.08%	−2.86%	−3.04%
		(4.40)	(6.10)	(5.21)	(4.35)	(4.49)	(6.02)	(6.12)	(4.72)
	2	−0.61%	−1.03%	−1.13%	−1.02%	−0.13%	−0.35%	−0.62%	−0.79%
		(3.82)	(4.30)	(4.29)	(2.49)	(0.80)	(1.13)	(1.60)	(1.89)
	3	−0.12%	−0.13%	−0.60%	−0.99%	−0.32%	−0.20%	−0.17%	−0.14%
		(0.91)	(0.45)	(1.48)	(1.88)	(2.28)	(0.77)	(0.58)	(0.33)
	4	−0.02%	0.10%	−0.14%	−0.07%	−0.12%	−0.04%	0.01%	0.08%
		(0.13)	(0.33)	(0.44)	(0.17)	(0.60)	(0.13)	(0.02)	(0.29)
	5	−0.21%	−0.24%	−0.08%	0.15%	−0.07%	−0.05%	−0.35%	−0.06%
		(1.44)	(0.95)	(0.23)	(0.29)	(0.39)	(0.18)	(1.01)	(0.12)
	6	−0.02%	0.13%	0.21%	0.43%	−0.08%	0.05%	0.22%	−0.09%
		(0.11)	(0.39)	(0.46)	(0.91)	(0.40)	(0.17)	(0.55)	(0.17)
	7	−0.03%	−0.06%	−0.18%	−0.25%	0.03%	0.02%	−0.16%	−0.39%
		(0.17)	(0.23)	(0.48)	(0.45)	(0.17)	(0.09)	(0.55)	(1.02)
	8	0.53%	0.99%	1.09%	0.97%	0.19%	0.24%	0.46%	0.49%
		(2.82)	(3.38)	(3.08)	(2.51)	(1.09)	(0.86)	(1.31)	(1.44)
	9	0.41%	0.75%	1.19%	1.28%	0.41%	0.65%	0.85%	0.75%
		(2.30)	(2.24)	(2.51)	(2.57)	(3.04)	(2.10)	(2.21)	(1.92)
	10	1.05%	1.65%	2.56%	2.87%	1.01%	1.75%	2.61%	3.19%
		(3.08)	(2.65)	(2.85)	(2.62)	(2.87)	(3.00)	(3.26)	(3.13)
	P10−P1	2.04%	3.89%	5.59%	6.39%	1.96%	3.83%	5.47%	6.24%
		(4.28)	(5.15)	(4.69)	(4.33)	(4.35)	(5.49)	(5.57)	(4.96)

(b) Table 4, Part 2: Size and book-to-market-adjusted returns

Panel C: Size, book-to-market and momentum-adjusted returns									
Decile	1	−0.57%	−1.66%	−2.31%	−2.62%	−0.55%	−1.48%	−2.19%	−2.21%
		(2.73)	(5.02)	(4.51)	(3.77)	(2.82)	(4.70)	(5.29)	(3.90)
	2	−0.49%	−0.76%	−0.79%	−0.62%	−0.03%	−0.20%	−0.36%	−0.60%
		(3.19)	(3.51)	(3.60)	(1.92)	(0.21)	(0.68)	(1.01)	(1.65)
	3	−0.05%	−0.07%	−0.63%	−0.93%	−0.22%	−0.07%	−0.01%	0.11%
		(0.47)	(0.28)	(1.77)	(2.02)	(1.63)	(0.30)	(0.02)	(0.29)
	4	0.04%	0.27%	0.07%	0.17%	0.03%	0.15%	0.14%	0.32%
		(0.25)	(1.05)	(0.29)	(0.47)	(0.20)	(0.65)	(0.55)	(1.14)
	5	−0.23%	−0.34%	−0.22%	0.06%	−0.13%	−0.20%	−0.52%	−0.31%
		(1.90)	(1.55)	(0.74)	(0.15)	(0.89)	(0.87)	(1.92)	(0.87)
	6	−0.01%	0.08%	0.10%	0.25%	−0.09%	0.02%	0.12%	−0.07%
		(0.08)	(0.26)	(0.24)	(0.58)	(0.52)	(0.07)	(0.31)	(0.14)
	7	−0.04%	−0.11%	−0.17%	−0.35%	−0.04%	−0.07%	−0.32%	−0.62%
		(0.25)	(0.42)	(0.54)	(0.69)	(0.25)	(0.36)	(1.18)	(1.94)
	8	0.45%	0.85%	0.89%	0.84%	0.19%	0.11%	0.31%	0.36%
		(2.82)	(3.26)	(2.56)	(2.24)	(1.23)	(0.45)	(0.93)	(1.24)
	9	0.17%	0.42%	0.81%	0.78%	0.22%	0.44%	0.67%	0.45%
		(0.99)	(1.35)	(1.82)	(1.78)	(1.74)	(1.67)	(1.91)	(1.22)
	10	0.71%	1.26%	2.18%	2.32%	0.62%	1.30%	2.16%	2.58%
		(2.50)	(2.39)	(2.92)	(2.54)	(2.02)	(2.53)	(3.21)	(3.06)
	P10−P1	1.28%	2.92%	4.49%	4.95%	1.17%	2.78%	4.35%	4.78%
		(3.13)	(4.26)	(4.45)	(3.93)	(2.77)	(4.07)	(4.90)	(4.28)

(a) Table 4, Part 3: Momentum-adjusted returns

Table 5

Returns to portfolio strategies based on ΔBREADTH, disaggregated by size

The sample includes stocks from NYSE/AMEX and NASDAQ between 1979 and 1998. In each quarter t, for stocks in size quintiles, residual $\Delta BREADTH_t$ is formed by regressing $\Delta BREADTH_t$ on $\Delta HOLD_t$, the change in aggregate holdings of mutual funds in that quarter. Stocks are ranked (into deciles) relative to other stocks in their size quintile on the basis of residual $\Delta BREADTH_t$. Then for stocks in similar deciles of residual $\Delta BREADTH_t$, an equal-weighted portfolio is formed and the performance is tracked over four quarters. This table reports the average returns of the portfolios in deciles 1 and 10 along with the difference in the returns of portfolios in deciles 10 and 1, P10–P1. Panels A, B, and C present these results using raw returns, size/book-to-market adjusted returns, and size/book-to-market/momentum adjusted returns, respectively. t-statistics, which are in parentheses, are adjusted for serial-correlation using a Newey-West estimator with lags of up to four quarters.

Cumulative returns after	Quintile 1		Quintile 2		Quintile 3		Quintile 4		Quintile 5	
	2 Quarters	4 Quarters	2 Quarters	4 Quarters	2 Quarters	4 Quarters	2 Quarters	4 Quarters	2 Quarters	4 Quarters
Panel A: Raw returns, sort on residual ΔBREADTH										
Decile 1	8.48%	19.45%	5.32%	14.56%	5.40%	11.62%	7.00%	15.29%	7.93%	15.87%
	(3.67)	(4.39)	(2.68)	(3.81)	(3.07)	(3.98)	(3.72)	(4.94)	(5.36)	(5.42)
10	9.09%	18.64%	10.46%	20.96%	9.67%	19.86%	9.24%	20.05%	9.49%	20.71%
	(3.93)	(4.08)	(4.25)	(4.62)	(4.82)	(5.66)	(4.53)	(5.10)	(4.90)	(6.17)
P10−P1	1.22%	−1.07%	5.15%	6.40%	4.27%	8.24%	2.24%	4.76%	1.56%	4.84%
	(1.57)	(0.71)	(5.88)	(3.36)	(3.81)	(5.31)	(1.96)	(2.49)	(1.45)	(3.18)
Panel B: Size and book-to-market-adjusted returns, sort on residual ΔBREADTH										
Decile 1	−1.27%	−0.61%	−2.61%	−2.77%	−2.73%	−5.06%	−1.13%	−2.06%	−0.76%	−1.79%
	(2.13)	(0.65)	(5.72)	(2.47)	(4.08)	(5.40)	(1.50)	(2.08)	(1.36)	(1.96)
10	−0.06%	−1.49%	2.52%	3.50%	1.53%	3.10%	1.40%	3.23%	0.90%	2.83%
	(0.11)	(1.44)	(3.34)	(2.87)	(2.16)	(2.41)	(1.57)	(1.94)	(1.25)	(2.79)
	1.22%	−0.86%	5.13%	6.27%	4.26%	8.15%	2.53%	5.28%	1.66%	4.62%
P10−P1	(1.69)	(0.60)	(6.40)	(3.58)	(3.94)	(5.51)	(2.34)	(2.88)	(1.52)	(3.12)
Panel C: Size, book-to-market, and momentum-adjusted returns, sort on residual ΔBREADTH										
Decile 1	−1.12%	−0.74%	−1.96%	−2.01%	−1.90%	−3.79%	−0.62%	−1.20%	−0.53%	−1.44%
	(2.09)	(0.85)	(5.16)	(2.00)	(2.88)	(4.20)	(0.88)	(1.23)	(0.94)	(1.61)
10	−0.27%	−1.42%	2.01%	3.00%	1.17%	2.44%	1.06%	2.63%	0.28%	1.89%
	(0.52)	(1.54)	(2.87)	(2.81)	(1.85)	(2.21)	(1.48)	(1.89)	(0.49)	(2.33)
P10−P1	0.87%	−0.66%	3.97%	5.02%	3.07%	6.23%	1.68%	3.82%	0.80%	3.33%
	(1.35)	(0.51)	(5.03)	(3.14)	(2.86)	(4.43)	(1.95)	(2.44)	(0.87)	(2.66)

(b) Table 5: Returns disaggregated by size

7.4 Table 5: Returns to portfolio strategies by size

Read:

- Table 5 repeats the portfolio exercise by size quintile.
- The predictive power of residual $\Delta BREADTH$ is present across size classes but is generally stronger for smaller and mid-sized stocks than for the very largest stocks.
- Momentum-adjusted results remain positive in several size groups, although significance is weaker in some subsamples.

Detailed interpretation: Size disaggregation matters because breadth is naturally higher for large stocks and because arbitrage constraints are likely stronger in smaller stocks. The table shows that the effect is not confined to one size class, but the economic strength is more visible where arbitrage is plausibly more limited.

Economic message: The breadth effect behaves like a limits-to-arbitrage signal: it is strongest where constrained pessimism should matter most.

7.5 Table 6: Forecasting returns with $\Delta BREADTH$: Fama–MacBeth regressions

Read:

- Table 6 estimates quarterly cross-sectional return regressions for one-, two-, three-, and four-quarter horizons.
- The $\Delta BREADTH$ coefficient is positive and statistically significant at each horizon.
- At the four-quarter horizon, the coefficient remains economically large after including $\Delta HOLD$, size, book-to-market, momentum, and turnover.
- $\Delta HOLD$ is positive in simple specifications but becomes much weaker when included alongside $\Delta BREADTH$ and other controls.

Detailed interpretation: This regression evidence complements the portfolio sorts. The main result is not created by decile cutoffs or portfolio formation rules. At the individual-stock level, a larger increase in breadth forecasts higher future returns.

Economic message: Breadth contains incremental return-predictive information beyond aggregate mutual-fund demand and standard characteristics.

Table 6
Forecasting returns with ΔBREADTH: Fama-MacBeth regressions
The sample includes stocks from the NYSE, AMEX, and NASDAQ between 1979 and 1998 with a market capitalization above the 20th percentile using NYSE breakpoints. The dependent variables are raw returns over one to four quarters. $\Delta BREADTH_t$ is the change in the breadth of ownership for a stock in quarter t. $\Delta HOLD_t$ is the change in aggregate mutual fund holdings of a stock in quarter t. $LOGSIZE_t$ is the log of market capitalization at the end of quarter t. BK/MKT_t is the most recently available observation of book-to-market ratio at the end of quarter t. MOM12 is the raw return from the beginning of quarter $t-3$ to the end of quarter t. $XTURNOVER_t$ is share turnover demeaned within each quarter by the average turnover for the firm's exchange (either NYSE/AMEX or NASDAQ). t-statistics, which are in parentheses, are adjusted for serial correlation and heteroskedasticity.

	1. $\Delta BREADTH_t$ only	2. $\Delta HOLD_t$ only	3. $\Delta BREADTH_t$ and $\Delta HOLD_t$	4. Additional controls
Panel A: Raw returns over one quarter				
$\Delta BREADTH_t$	2.045		2.055	1.187
	(3.72)		(3.49)	(2.67)
$\Delta HOLD_t$		0.257	0.093	0.072
		(2.90)	(1.25)	(1.17)
$LOGSIZE_t$				−0.003
				(1.01)
BK/MKT_t				0.008
				(1.71)
$MOM12_t$				0.029
				(4.02)
$XTURNOVER_t$				−0.038
				(2.49)
No. of quarters	79	79	79	79
Average R^2	1.2%	0.8%	1.9%	9.5%
Panel B: Raw returns over two quarters				
$\Delta BREADTH_t$	3.502		3.552	2.022
	(4.29)		(4.04)	(2.76)
$\Delta HOLD_t$		0.430	0.119	0.129
		(3.54)	(1.22)	(1.45)
$LOGSIZE_t$				−0.001
				(0.25)
BK/MKT_t				0.014
				(1.82)
$MOM12_t$				0.059
				(4.62)
$XTURNOVER_t$				−0.071
				(2.50)
No. of quarters	79	79	79	79
Average R^2	1.2%	0.7%	1.9%	10.4%
Panel C: Raw returns over three quarters				
$\Delta BREADTH_t$	4.403		4.536	2.803
	(4.94)		(4.83)	(3.10)
$\Delta HOLD_t$		0.592	0.154	0.132
		(3.66)	(1.20)	(1.15)

(a) Table 6, Part 1: One- to three-quarter Fama–MacBeth results

Table 6 (continued)

	1. $\Delta BREADTH_t$ only	2. $\Delta HOLD_t$ only	3. $\Delta BREADTH_t$ and $\Delta HOLD_t$	4. Additional controls
$LOGSIZE_t$				0.007
				(1.04)
BK/MKT_t				0.022
				(2.38)
$MOM12_t$				0.076
				(4.35)
$XTURNOVER_t$				−0.109
				(3.09)
No. of quarters	79	79	79	79
Average R^2	1.2%	0.8%	1.9%	10.1%
Panel D: Raw returns over four quarters				
$\Delta BREADTH_t$	4.469		4.504	2.932
	(3.51)		(3.77)	(3.18)
$\Delta HOLD_t$		0.722	0.207	0.145
		(3.25)	(1.13)	(0.95)
$LOGSIZE_t$				0.013
				(1.64)
BK/MKT_t				0.029
				(2.60)
$MOM12_t$				0.084
				(4.16)
$XTURNOVER_t$				−0.141
				(3.10)
No. of quarters	79	79	79	79
Average R^2	1.2%	0.9%	1.9%	9.9%

regression, a two-standard deviation spread in ΔBREADTH generates a differential in expected returns of 2.71% over a four-quarter horizon.

To put the forecasting power of ΔBREADTH in perspective, consider the

(b) Table 6, Part 2: Four-quarter Fama–MacBeth results

7.6 Table 7: Forecasting returns with $\Delta BREADTH$: four-quarter Fama–MacBeth robustness checks

Read:

- Table 7 begins from the baseline four-quarter Fama–MacBeth specification.
- The breadth coefficient remains positive when momentum is lagged by one month.
- Decomposing momentum into four three-month components does not eliminate the breadth effect.
- Decomposing $\Delta BREADTH$ into IN and OUT gives predicted signs: more new entrants predicts higher returns; more exits predicts lower returns.
- Adding future $\Delta HOLD$ reduces concerns that breadth simply forecasts future mutual-fund demand or price pressure.

Detailed interpretation: Table 7 addresses the most important alternative explanations. The result survives more detailed momentum controls, and the IN/OUT decomposition matches the model's logic. The future demand test is important because it helps distinguish valuation information from mechanical buying pressure.

Economic message: The robustness evidence strengthens the interpretation that breadth captures information withheld by short-sale constraints, not merely fund-flow demand.

Table 7
Forecasting returns with ΔBREADTH: four-quarter Fama-MacBeth robustness checks
The sample includes stocks from the NYSE, AMEX, and NASDAQ between 1979 and 1998 (1979–1997 for column 5) with a market capitalization above the 20th percentile using NYSE breakpoints. The dependent variables are raw returns over four quarters. $\Delta\text{BREADTH}_t$ is the change in the breadth of ownership for a stock in quarter t. IN_t is the fraction of mutual funds in the sample at both quarters $t-1$ and t that have established a new position in a stock at quarter t. OUT_t is the fraction of mutual funds that have completely removed an existing position in a stock at quarter t. ΔHOLD_t is the change in aggregate mutual fund holdings of a stock in quarter t. LOGSIZE_t is the log of market capitalization at the end of quarter t. BK/MKT_t is the most recently available observation of book-to-market ratio at the end of quarter t. MOM12 is the raw return from the beginning of quarter $t-3$ to the end of quarter t. MOM3 is the raw return from the beginning of quarter t to the end of quarter t. Column 2 uses MOM12 lagged one month. XTURNOVER_t is share turnover demeaned within each quarter by the average turnover for the firm's exchange (either NYSE/AMEX or NASDAQ). t-statistics, which are in parentheses, are adjusted for serial correlation and heteroskedasticity.

	1. Base-case	2. Momentum with one-month lag	3. Momentum decomposed	4. ΔBREADTH decomposed	5. With future ΔHOLD
$\Delta\text{BREADTH}_t$	2.932	2.486	2.390		2.336
	(3.18)	(2.29)	(2.36)		(2.10)
IN_t				3.674	
				(1.83)	
OUT_t				−2.008	
				(1.43)	
ΔHOLD_t	0.145	0.090	0.075	0.134	0.236
	(0.95)	(0.57)	(0.44)	(0.87)	(1.43)
ΔHOLD_{t+2}					1.832
					(6.37)
ΔHOLD_{t+1}					2.342
					(7.76)
ΔHOLD_{t+3}					2.274
					(8.05)
ΔHOLD_{t+4}					1.980
					(6.67)
LOGSIZE_t	0.013	0.019	0.002	0.013	0.007
	(1.64)	(2.54)	(0.31)	(1.41)	(0.96)
BK/MKT_t	0.029	0.030	0.022	0.028	0.027
	(2.60)	(2.49)	(1.93)	(2.39)	(2.70)
MOM12_t (MOM3_t	0.084	0.083	0.149	0.086	0.097
in column 3)	(4.16)	(4.61)	(4.66)	(4.50)	(4.68)
MOM3_{t-1}			0.141		
			(5.10)		
MOM3_{t-2}			0.075		
			(2.82)		
MOM3_{t-3}			0.016		
			(0.58)		
XTURNOVER_t	−0.141	−0.126	−0.128	−0.138	−0.133
	(3.10)	(2.57)	(2.92)	(2.50)	(3.00)
No. of Quarters	79	79	79	79	75
Average R^2	9.9%	10.1%	13.4%	10.8%	18.1%

Table 7: Four-quarter Fama–MacBeth robustness checks

8 Detailed result map

- **Result 1: Breadth is a better theoretical proxy than short interest.** Low short interest can mean either little pessimism or high shorting cost. Low breadth directly means fewer constrained investors are willing to be long.
- **Result 2: Breadth changes forecast returns.** Stocks with falling breadth underperform stocks with rising breadth.
- **Result 3: Breadth is linked to momentum.** The strong relation between $\Delta BREADTH$ and prior returns suggests that momentum is partly connected to short-sale constraints.
- **Result 4: Predictive power survives controls.** Characteristic adjustments and Fama–MacBeth controls reduce but do not eliminate the effect.
- **Result 5: The mechanism appears on both entry and exit margins.** New mutual fund entry and old mutual fund exit both move in the direction predicted by the theory.
- **Result 6: Breadth is not just future fund demand.** Adding future $\Delta HOLD$ does not eliminate the breadth effect.

9 Short summary

- The paper develops a model with differences of opinion and short-sale constraints.
- Low breadth of ownership means that many constrained investors are not willing to hold the stock.
- The model predicts that breadth should be positively related to future risk-adjusted returns.
- Short interest is theoretically ambiguous as a predictor because it mixes disagreement with shorting costs and arbitrageur risk tolerance.
- The empirical measure of breadth comes from mutual-fund holdings from 1979 to 1998.
- The key variable is the quarterly change in breadth, $\Delta BREADTH$.
- Stocks with large declines in breadth subsequently underperform stocks with large increases in breadth.
- The return spread remains after adjustment for size, book-to-market, and momentum.
- Breadth is strongly related to momentum, suggesting that momentum may be connected to binding short-sale constraints.
- Fama–MacBeth regressions confirm the portfolio-sort evidence.
- Robustness checks address alternative momentum controls, future fund demand, seasonal effects, and outliers.
- The main limitation is that mutual funds are only one class of constrained investors, so breadth is an incomplete measure of the full market.

10 Conclusion

10.1 Core conclusion

Chen, Hong, and Stein provide evidence that short-sale constraints affect equilibrium stock prices. Their central empirical insight is that breadth of ownership reveals the extent to which constrained investors are unwilling to hold a stock. When breadth falls, pessimistic views are more likely to be excluded from price, and subsequent returns are lower.

10.2 Economic intuition

The intuition is simple. If shorting is difficult, a pessimistic investor cannot express a negative view by shorting the stock. The investor can only avoid holding it. Therefore, the number of investors willing to hold the stock is informative. A stock held by fewer mutual funds is more likely to have pessimists on the sidelines, and its price may reflect only optimistic valuations.

10.3 Incremental contribution in asset pricing

The paper moves the short-sale-constraints literature away from short interest and toward ownership breadth. This is important because short interest measures actual short positions, while the theory is about the investors who would like to short but do not. Breadth is closer to that missing counterfactual.

10.4 Implications for momentum

The paper's results suggest that momentum is not just a statistical regularity. When a stock has performed poorly, many mutual funds leave it, but they cannot short it. The price may remain too high because negative information is not fully incorporated. This connects the short side of momentum profits to institutional constraints.

10.5 Limits and caveats

The mutual-fund data do not cover every constrained investor. Breadth could partly reflect mutual-fund clienteles, preferences, or skill. The paper controls for aggregate fund holdings, but it cannot fully rule out all alternative interpretations. The design is also predictive rather than causal.

10.6 Takeaway

Breadth of ownership is a useful market signal because it measures participation by constrained investors. Declining breadth means more sidelined pessimism, tighter binding short-sale constraints, and lower future returns.